Extracting Social and Clinical Impacts of Substance Use from Reddit

CE7455 Project – Group G26

Ismail Elyamany Huang Chongtian Xia Yitong Zhu Ruijie

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Nanyang Technological University

The Task: Impact Extraction from Reddit

SMM4H–HeaRD 2026 Task 7 Contest Token-level NER on opioid-subreddit posts.

Task: To identify two entity types using BIO tagging:

- **ClinicalImpacts** – for e.g. overdose, withdrawal, depression, insomnia ...
- **SocialImpacts** – for e.g. job loss, arrest, homelessness, broken relationships ...

Example

> *"I lost my job last month and ever since I've had constant panic attacks."*

Motivation

Useful for population-level signal mining from user-generated text – public health surveillance, harm-reduction research.

Why It's Hard

1. **Train/Test Distribution Shift**
   - Train is 75% Clinical, 25% Social.
   - Test is ≈30% Clinical, 70% Social.
   - Models trained on Train over-predict Clinical.
2. **Implicit, Colloquial Language**
   - "those weeks were hell" (Clinical)
   - "I lost everything" (Social)
   - No clinical vocabulary, no named triggers.

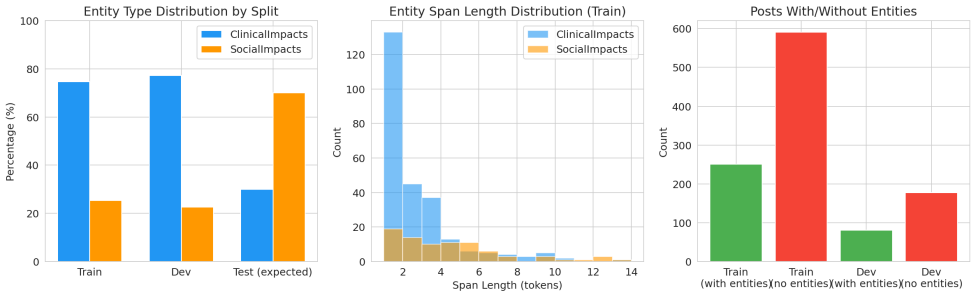
Reported Ceilings

System	Relaxed F1
Human expert (κ)	0.81
DeBERTa (published SOTA)	**0.610**
GPT-4o 3-shot	0.44

Even our best dev ensemble is still about 18 points below the human ceiling $\Rightarrow$ lots of headroom.

Dataset: Reddit-Impacts 2.0

1,378 opioid-subreddit posts with BIO annotations.



Left: Train/Dev are Clinical-heavy; Test flips to ≈70 % Social. **Middle:** Social spans are longer and heavier-tailed than Clinical. **Right:** most Dev posts contain no entities at all.

Preprocessing Technique: Raw vs preprocessed

raw runtime regime

- Original Reddit text goes straight to tokenization.
- Keeps usernames, URLs, mojibake, emoji, and punctuation noise.
- Stronger lexical fidelity, but noisier token boundaries.

preprocessing-enabled regime

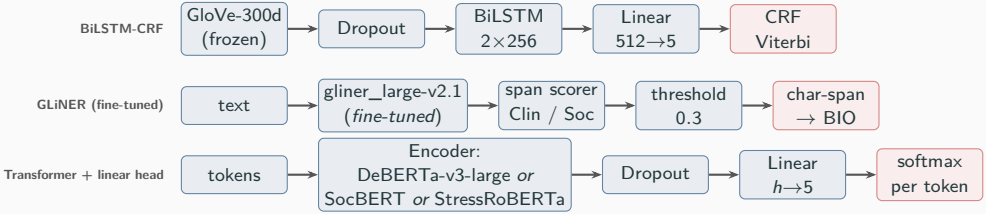
- Mojibake / broken-Unicode repair.
- `@user` and URL masking.
- Emoji / punctuation scrub before encoding.
- Masked positions restored as `O` tags at inference.

Example

Raw	`@john i couldn't keep a job !!! https://...`
Preprocessed	`@USER i couldn't keep a job`

Why this matters for NER. We want less lexical noise *without* breaking BIO alignment. That trade-off is mixed: preprocessing helps BiLSTM and GLiNER, but the plain DeBERTa baseline is actually stronger on raw text.

Model Comparison: Architectures and Backbones



Model	Raw	Prep
BiLSTM-CRF	0.251	0.296
GLiNER (fine-tuned)	0.463	0.524
SocBERT (Reddit-domain)	–	0.404
StressRoBERTa (mental-health)	–	0.482
DeBERTa-v3-large baseline	**0.595**	0.545

General-purpose DeBERTa still dominates; preprocessing helps BiLSTM and GLiNER but slightly hurts the plain DeBERTa baseline on relaxed F1. Domain-adapted backbones are only evaluated in the preprocessed regime, and zero-shot GLiNER still bottoms out at 0.320.

Evaluation Protocol Before the Innovations

Component	What we report
Relaxed F1	Correct entity type with span overlap; tolerant to boundary mismatch.
Strict F1	Correct entity type *and* exact span boundary match.
Model selection	Best learning rate is chosen per family, separately within each regime.
Comparison rule	Innovations are compared *within* raw or prep first; only the later ensemble slide mixes both regimes.
Data split	All results here are dev-set results; no test labels are used.

This is why the slides below show paired raw-vs-preprocessed numbers: preprocessing is treated as an experimental axis, not as an invisible implementation detail.

Our Strategy: Seven Contribution Buckets Beyond the Baselines

1. **Loss/prompt/multitask ablations** – focal loss, definition prompting, multi-task head, synthetic-data curriculum.
2. **Advanced regularization** – O-class downweighting (recall-boost), R-Drop, FGM + SWA.
3. **Preprocessing as an axis** – raw vs cleaned runtime regimes.
4. **Structured prediction pipelines** – hierarchical DeBERTa, two-step extract→classify, joint sentence+token heads, span-nested GLiNER.
5. **Domain-adapted backbones** – SocBERT (Reddit), StressRoBERTa (mental health).
6. **Weight-space model soup** – top-K checkpoint averaging.
7. **Prediction-space ensemble search** – exhaustive subsets, majority vote and probability averaging.

Each innovation is evaluated with a small LR sweep [2e-5, 5e-5] inside each regime (raw, preprocessed), and the best LR per family is reported.

Innovation 1: Distribution-Aware Focal Loss

Definition. Let p_t be the predicted probability of the *true* class for a token. Standard cross-entropy is $\mathrm{CE}(p_t) = -\log p_t$. Focal loss adds a modulating factor:

$$\mathcal{L}_{\text{focal}}(p_t) \;=\; -\,\alpha_t\,(1-p_t)^{\gamma}\,\log p_t$$

- $(1-p_t)^{\gamma}$ – **focusing term.** When the model is already confident ($p_t \to 1$), the factor $\to 0$, so easy tokens contribute almost nothing to the gradient. Hard tokens (p_t small) keep near-full loss. We use $\gamma = 2.0$.
- α_t – **per-class weight.** Lets us emphasize under-represented classes. We use $\alpha_{\text{Social}} = 1.5$, $\alpha_{\text{Clinical}} = \alpha_{\text{O}} = 1.0$ to push the model toward Social (which dominates the test set).

Regime	Baseline (R / S)	+ Focal (R / S)
raw	0.595 / 0.409	0.541 / 0.341
preprocessed	0.545 / 0.421	0.538 / 0.358

Finding: aggressive Social reweighting pushes the model to *over*-fire on Social tokens in both regimes. Preprocessing softens the strict-F1 drop a little, but focal loss remains below the plain baseline either way. Distribution shift is real, but class weighting is the wrong lever for it.

Innovation 2: Definition Prompting

Prepend natural-language *definitions* of both entity types before the post; mask them from the loss.

[CLS] *Clinical Impacts are physical or psychological consequences . . . overdose, withdrawal, . . . Social Impacts are social, occupational, legal . . . job loss, arrest, . . .* [SEP]

↓

`I lost my job last month and had constant panic attacks.` [SEP]

↓

Labels: `-100 -100 . . . -100 | O B-Soc I-Soc O O O B-Clin I-Clin I-Clin O`

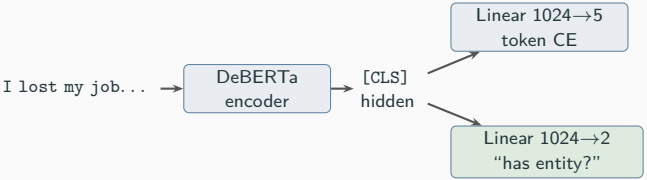
Mechanism: `-100` on prefix ⇒ definition tokens contribute *attention context* but no gradient. The encoder sees an explicit label ontology on every example.

Regime	Baseline (R / S)	+ Definitions (R / S)
raw	0.595 / 0.409	0.575 / 0.350
preprocessed	0.545 / 0.421	**0.613** / 0.410

Finding: definitions only really pay off once the text is cleaned. On raw they hurt; on preprocessed they become the best relaxed-F1 single-model recipe in the project.

Innovation 3: Multi-task Auxiliary Head

Share the DeBERTa encoder between token NER and a sentence-level *entity-presence* classifier.



Losses. $\mathcal{L} = \mathrm{CE}_{\text{token}} + 0.3 \cdot \mathrm{CE}_{\text{presence}}$, where the target for $\mathrm{CE}_{\text{presence}}$ is $y = \mathbf{1}[\text{sentence has any B/I tag}]$.

Regime	Baseline (R / S)	+ Multi-task (R / S)
raw	0.595 / 0.409	0.581 / 0.382
preprocessed	0.545 / 0.421	0.550 / 0.394

Finding. On raw, multi-task slightly hurts. On preprocessed it recovers to a small relaxed-F1 gain over the plain preprocessed baseline, but strict F1 still trails. The auxiliary head is more useful as a feature source for later recipes than as a standalone win.

Innovation 4: Recall Boost via O-class Downweighting

Diagnosis. Error analysis showed recall lagging precision: 29 missed spans, 13 of them single-token implicit expressions.

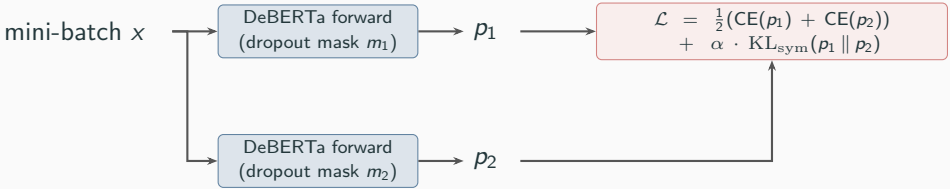
Intervention. Cross-entropy weight on the `O` class: $1.0 \rightarrow 0.2$. Entity classes keep weight 1.0 while Social classes increases to 1.5. Combined with multi-task.

Regime	Baseline (R / S)	Seed 42 (R / S)	Seed 123 (R / S)
raw	0.595 / 0.409	0.582 / 0.312	0.562 / 0.357
preprocessed	0.545 / 0.421	0.560 / 0.330	0.552 / 0.359

Effect: preprocessing lifts strict F1 a little for both seeds, but recall-boost still stays below its own regime baseline. The O-downweight continues to trade precision for recall. Not a blockbuster alone, *but* it produces errors that complement the other recipes – which matters for the ensemble.

Innovation 5: R-Drop Consistency Regularization

Two stochastic forward passes on the *same* batch, then force their distributions to agree.



$\alpha = 1.0$, symmetric KL over non-masked tokens only.

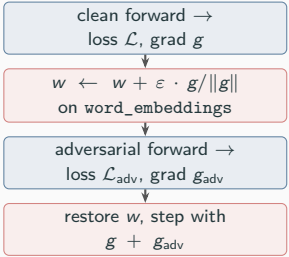
No new parameters – dropout itself is used as a regularizer that penalizes unstable predictions.

Regime	Baseline (R / S)	+ R-Drop s123 (R / S)
raw	0.595 / 0.409	0.561 / 0.360
preprocessed	0.545 / 0.421	0.577 / 0.380

Preprocessing is what makes R-Drop competitive on relaxed F1: the preprocessed run reaches 0.577 vs 0.561 on raw text, although strict F1 still stays below the plain preprocessed baseline.

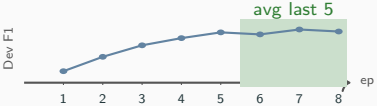
Innovation 6: FGM Adversarial + SWA

FGM (per training step)



$\varepsilon = 0.5$; perturbation lives only on the embedding layer.

SWA (stochastic weight averaging)

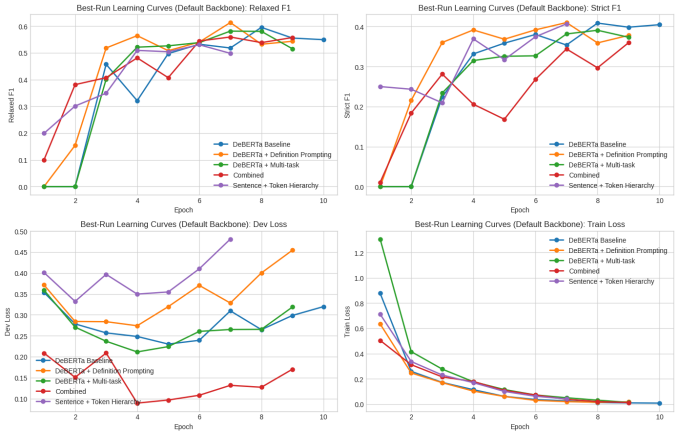


After epoch 10, running mean of `state_dict` over remaining epochs → final checkpoint. Smooths end-of-training oscillation.

Regime	Baseline (R / S)	+ FGM + SWA (R / S)
raw	0.595 / 0.409	0.573 / 0.356
preprocessed	0.545 / 0.421	0.547 / 0.256

Robustness recipe: slightly below the raw baseline, and substantially worse after preprocessing. This one helps mostly through diversity, not standalone dev performance.

Training Dynamics Across Recipes



Top row: Definition Prompting (orange) tops both Relaxed and Strict F1 from epoch 4 onward – the clearest single-recipe win. **Bottom row:** Dev loss separates recipes even when F1 curves look close; Sentence+Token Hierarchy drifts up after epoch 6, signalling mild overfit.

Structured Prediction Pipelines

Four structured variants on top of DeBERTa, each testing a different inductive bias:

Pipeline	Idea	raw (R / S)	preprocessed (R / S)
Hierarchical DeBERTa	`[CLS]` sentence classifier gates a multi-task NER; 10 % of no-impact rows kept	**0.604** / 0.372	0.559 / 0.383
Two-Step Impact Pipeline	Stage 1: binary span extraction; Stage 2: clinical / social classifier per span	–	0.534 / 0.404
Sentence + Token Hierarchy	One model with joint sentence presence head and token BIO head	–	0.530 / 0.375
Span-nested GLiNER	Span scorer with nested-span enumeration and overlap-aware decoding	–	0.443 / 0.491

Hierarchical DeBERTa is the clear relaxed-F1 winner and anchors the winning ensemble on the next slide. Its strict F1 actually improves slightly under preprocessed, even though relaxed F1 drops. The other structured variants are only present in the notebook under the preprocessed regime.

Model soup (weight-space aggregation). Orthogonal to everything above: average top-K dev-F1 checkpoints from a single LR sweep ($\bar{\theta} = \frac{1}{K}\sum_k \theta_k$). Best soup: recall-boost s42 top-3 at 0.595 (strict 0.331), matching the plain baseline on relaxed F1 but still below definition prompting. Soup is only defined for same-architecture runs – GLiNER, BiLSTM, and every pipeline family above are *not* soupable.

Innovation 7: Ensemble Search – Majority Vote Beats SOTA

Two complementary searches, both exhaustive over size-2–5 subsets:

- **Majority vote, cross-version:** top-7 relaxed + top-7 strict from each regime (wihtout preprocessing / with preprocessing) ⇒ 28 candidates, **122,409 combinations**. Operates on BIO tag sequences – every family is eligible.
- **Probability average** 7 candidates, 112 combinations. Averages per-token class distributions, so GLiNER span models and structured pipelines are excluded.

Ensemble	Relaxed F1	Strict F1	Precision	Recall
2 models (majority)	0.623	0.465	0.586	0.664
3 models (majority)	0.621	0.507	0.638	0.605
4 models (majority)	**0.635**	0.509	0.648	0.623
5 models (majority)	0.631	**0.518**	0.670	0.597
Best probability average (raw, 7 cand.)	0.614	0.431	0.653	0.580

Best ensemble: hierarchical-pipeline anchor + 3 complementary DeBERTa recipes across both regimes. **Beats published SOTA 0.610 on dev** (+0.025). Hard voting > soft averaging here (0.635 vs 0.614).

Full Ablation Table (LR-Selected per Family)

Model	Relaxed F1	Strict F1	Best LR	Epoch
BiLSTM-CRF	0.296	0.256	5e-4	9
GLiNER (fine-tuned)	0.524	–	2e-5	–
Span-nested GLiNER	0.443	–	1e-5	–
SocBERT baseline	0.404	0.277	5e-5	–
StressRoBERTa baseline	0.482	0.270	2e-5	–
DeBERTa baseline	0.595	0.409	2e-5	8
+ Focal loss	0.541	0.341	2e-5	8
+ Multi-task	0.581	0.382	5e-5	7
+ Definition prompting	**0.613**	0.410	5e-5	7
Hierarchical DeBERTa pipeline	0.604	0.372	2e-5	–
Two-Step Impact Pipeline	0.534	0.404	2e-5	–
Sentence + Token Hierarchy	0.530	0.374	5e-5	–
Recall-Boost (seed 42)	0.582	0.312	2e-5	8
R-Drop (seed 123)	0.577	0.380	2e-5	9
FGM + SWA (seed 42)	0.573	0.356	2e-5	12
Model soup (recall-boost s42)	0.595	0.331	2e-5	–
4-Model Ensemble (majority vote)	**0.635**	0.509	–	–
GPT-4o 3-shot (reported)	*0.440*	–	–	–
DeBERTa (published SOTA)	*0.610*	–	–	–

Dev-set evaluation. Each family has its own LR sweep; the row reports the best LR. The strict-optimal majority-vote ensemble is a different 5-model subset at 0.518 strict F1.

Error Analysis on the Best Relaxed Single Model

Definition Prompting (lr 5e-5), Relaxed F1 $= 0.613$, Strict F1 $= 0.410$.

Error categories (Dev)

- **Boundary errors** dominate – Relaxed/Strict gap of ≈ 0.20.
- Missed spans concentrated on single-token implicit expressions.
- Long multi-word Social spans get partial matches.
- Type confusion is rare.

Qualitative observations

- Implicit: "those weeks were hell" often missed.
- "charged with disorderly conduct" $\rightarrow$ partial span.
- Negated phrases correctly rejected.
- Third-person references are mostly ignored.

Boundary errors and implicit false negatives are the two buckets that would pay off most for future work.

Three Key Insights

1. **LR sweeps are non-negotiable.** Best LR is family-specific (2e-5 for baseline, 5e-5 for definition prompting and multi-task); a single global LR masks the true ordering between recipes and flips several conclusions.
2. **Training-objective diversity beats individual model polish.** Single models plateau around 0.60; a 4-model majority-vote ensemble climbs to **0.635** – a +0.022 jump over the best relaxed single model from combining *different recipes* (hierarchical, definition, recall-boost, R-Drop), not more seeds of the same one.
3. **Majority vote beats probability averaging.** On this small, noisy dev set, hard voting (**0.635**) is more robust than soft averaging (0.614). Confident-but-occasionally-wrong models don't combine well via logits.

Limitations & Future Work

Limitations

- Dev-set only – no test-set access; numbers will likely shift under the real 70 % Social distribution.
- Our ensemble beats the published SOTA (0.635 vs 0.610) on dev, but 258 samples is a noisy reference – the gap needs test-set confirmation.
- Single-sentence inputs – no cross-post user-history context.

Future work

- LLM-synthesized data (Claude / GPT-4o) targeting implicit Social cases.
- Greedy forward selection for principled ensemble membership.
- Cross-domain transfer from biomedical NER corpora.
- Instruction-tuned Llama-3 with task-specific prompts.

Summary

- **Task:** BIO NER for Clinical/Social impacts of substance use on Reddit.
- **Models:** BiLSTM-CRF, GLiNER (fine-tuned), SocBERT, StressRoBERTa, DeBERTa-v3-large.
- **7 contribution buckets:** ablations, advanced regularization, preprocessing axis, structured pipelines, domain backbones, model soup, ensemble search – each evaluated under an LR sweep.
- **Best relaxed single model:** Definition Prompting, $F_1 = \mathbf{0.613}$ (exceeds baseline by $+0.018$).
- **Best system:** 4-model majority-vote ensemble, $F_1 = \mathbf{0.635}$, beating the published SOTA **0.610** on dev ($+0.025$).
- **Main takeaway:** training-objective diversity + majority voting $>$ any single-model trick we tried.